

DEEP LEARNING PROJECT REPORT

Spam Detection Using LSTM

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ABSTRACT

This project focuses on detecting spam messages using Deep Learning techniques. A Long Short-Term Memory (LSTM) model is trained on text data to classify messages as spam or not spam. The model is integrated into a Streamlit application where users can enter a message and receive instant predictions. This project demonstrates the use of Natural Language Processing (NLP) and deep learning in real-world applications.

INTRODUCTION

Spam message detection is an important application of Artificial Intelligence, especially in Natural Language Processing. This project aims to classify text messages as spam or not spam using an LSTM model.

The application is built using Python and Streamlit, allowing users to input a message and get real-time predictions.

OBJECTIVES

- To build a deep learning model using LSTM
- To classify messages as spam or not spam
- To create an interactive web application
- To provide real-time prediction

TOOLS AND TECHNOLOGIES

- Programming Language: Python
- Libraries: TensorFlow/Keras, NumPy, Joblib
- NLP Tools: Tokenizer, Padding
- Web Framework: Streamlit
- Model: LSTM (Recurrent Neural Network)

DATASET

The model is trained on a dataset containing SMS messages labeled as spam or not spam.

- Input: Text messages
- Output: Spam / Not Spam

HOW THE MODEL WORKS

- Text messages are cleaned and preprocessed
- Tokenization converts words into sequences of numbers
- Sequences are padded to fixed length (100)
- LSTM model learns patterns in text
- Model predicts whether message is spam or not

HOW THE APP WORKS

From your app:

1. User enters a message
2. Text is converted into sequences using tokenizer
3. Data is padded to fixed length
4. Model processes the input
5. Prediction is generated
6. Output is displayed (Spam / Not Spam)

CODE

```
RNN_app.py
1  import streamlit as st
2  import joblib
3  from tensorflow.keras.models import load_model
4  from tensorflow.keras.preprocessing.sequence import pad_sequences
5
6  st.title("Spam Detection App (LSTM)")
7
8  # Load model
9  model = load_model("spam_model.keras")
10
11 # Load tokenizer
12 tokenizer = joblib.load("tokenizer.joblib")
13
14 # Input
15 message = st.text_area("Enter your message")
16
17 # Predict button
18 if st.button("Predict"):
19     if message.strip() == "":
20         st.warning("Please enter a message")
21     else:
22         # Convert text → numbers
23         seq = tokenizer.texts_to_sequences([message])
24         padded = pad_sequences(seq, maxlen=100)
25
26         # Prediction
27         prediction = model.predict(padded)
28
29         # Output
30         if prediction[0][0] > 0.5:
31             st.error("Spam Message")
32         else:
33             st.success("Not Spam")
```

OUTPUT

Spam Detection App (LSTM)

Enter your message

"Hey, are we still meeting for lunch today?"

Predict

Not Spam

Spam Detection App (LSTM)

Enter your message

"Congratulations! You have won a free iPhone. Click here to claim now!"

Predict

Spam Message

RESULTS

From the experimental results:

- "Congratulations! You won a free prize" → Spam Message
- "Call me when you are free" → Not Spam

Model correctly classifies messages.

CONCLUSION

This project successfully demonstrates spam detection using LSTM. The model effectively identifies spam messages, and the Streamlit application provides an easy-to-use interface. The system is efficient and useful for real-world applications.